micro:bit

Introduction to Computer Science

Unit 8: Booleans

Microsoft

Lesson A
Understanding Booleans

What we'll learn in Lesson A

- Booleans
- Boolean operators

What is a Boolean?

Something that only has two values:

- True
- False

Booleans are used in programming for decision making, often deciding when certain functions and parts of programs should start or stop running.

What are some other examples of Booleans in daily life?

A coin is a type of Boolean. When you flip it, it only has two values: Heads or Tails.

Boolean operators

Like Math Operators (+, –, etc.), Booleans also have Operators:

AND

(Boolean value A is true) AND (Boolean value B is true)

Both A and B must be true

OR

(Boolean value A is true) OR (Boolean value B is true)

Either A or B can be true

NOT

(Boolean value A)

Returns the opposite of A

George Boole

November 2, 1815 – December 8, 1864

English mathematician, educator, philosopher, and logician. George worked in the fields of differential equations and algebraic logic, and is best known as the author of *The Laws of Thought* (1854) which contains Boolean algebra.

Two heads are better than one

1. Working in pairs, create a table of possible outcomes when two coins are flipped.
2. Then, flip two coins and record the outcomes in your table.
3. Now, imagine a game where Player A wins when the coins are the same, and Player B wins when the coins are different.
4. Write the pseudocode for each outcome using the Boolean operator **AND** where:
   - Heads = True
   - Tails = False

Microsoft

Two Heads Pseudocode

Your pseudocode might look like:

Coin A	Coin B	Results
True (Heads)	True (Heads)	If Coin A is true AND Coin B is true, add one to Player A score
True (Heads)	False (Tails)	If Coin A is true AND Coin B is false, add one to Player B score
False (Tails)	True (Heads)	If Coin A is false AND Coin B is true, add one to Player B score
False (Tails)	False (Tails)	If Coin A is false AND Coin B is false, add one to Player A score

What we learned in Lesson A

- Booleans and Boolean operators

Next time: Lesson B

- Code a double coin flipper
- Assess our learning with a quiz

Lesson B

Code a double coin flipper

What we'll learn in Lesson B

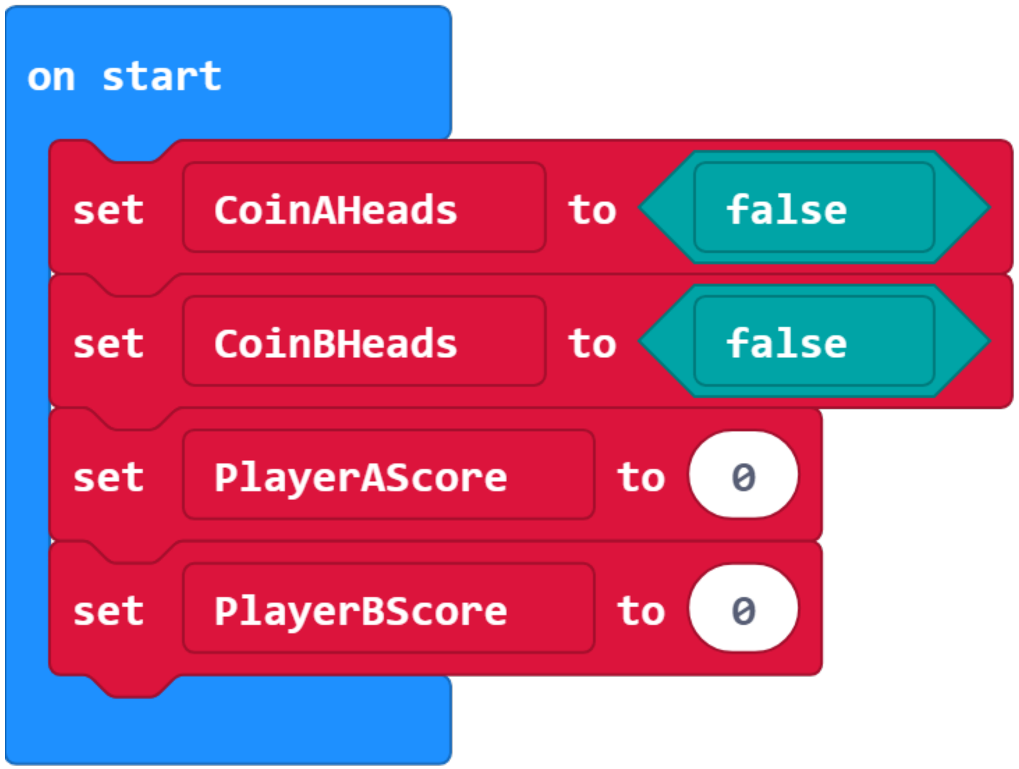
- Use Booleans to code a random double coin flipper
- Review how much we've already learned

Double coin flipper

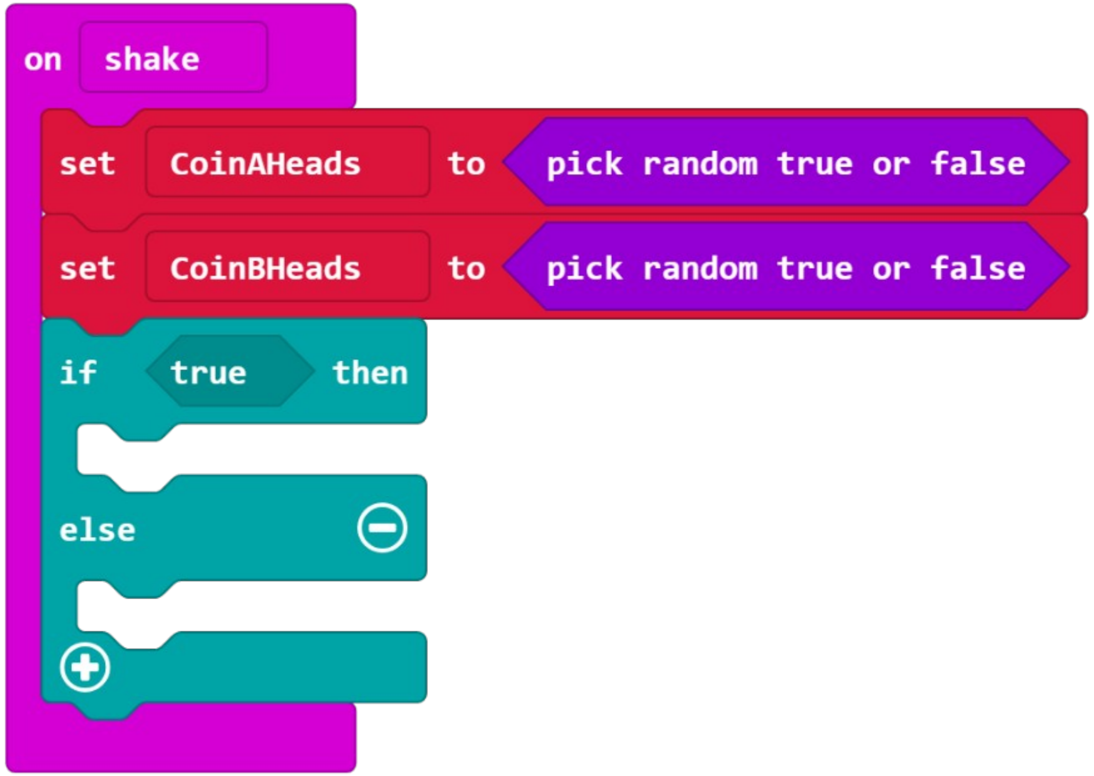
Use your pseudocode from the previous lesson to code this game.

Make the following variables:

- Boolean **CoinAHeads**:
  Is coin A heads? (True or false)
- Boolean **CoinBHeads**:
  Is coin B heads? (True or false)
- Number **PlayerAScore**:
  Tracks player A's wins
- Number **PlayerBScore**:
  Tracks player B's wins

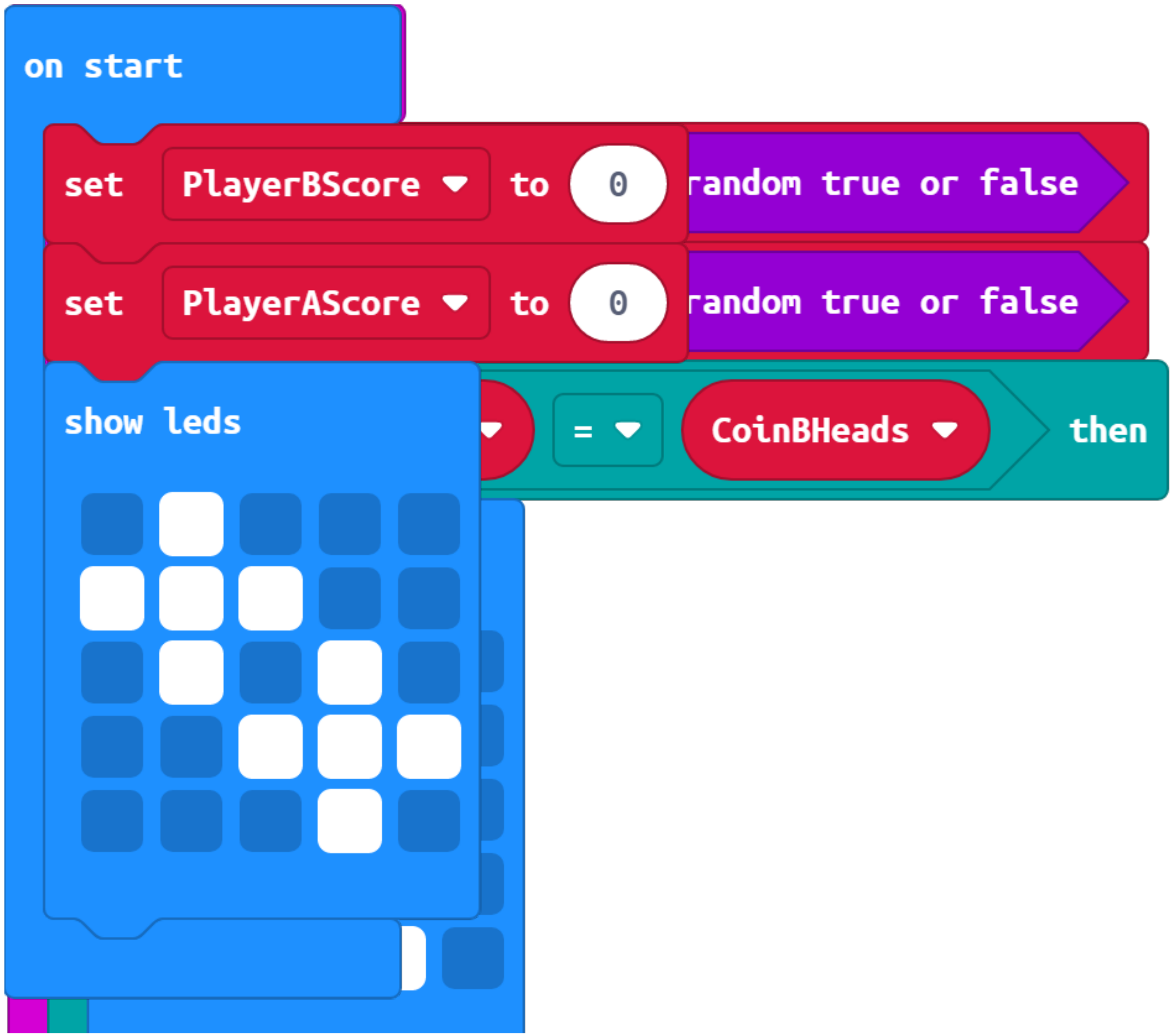


- Use an **On Shake** block to simulate the coin toss.
- Instead of setting **CoinAHeads** and **CoinBHeads** to False, use a **Pick Random True or False block**
- Use a conditional **IF… THEN… ELSE…** block to evaluate the outcome of the coin toss and increment the appropriate player’s score.
- Use a **Button** to display the current players’ scores.



Your final code

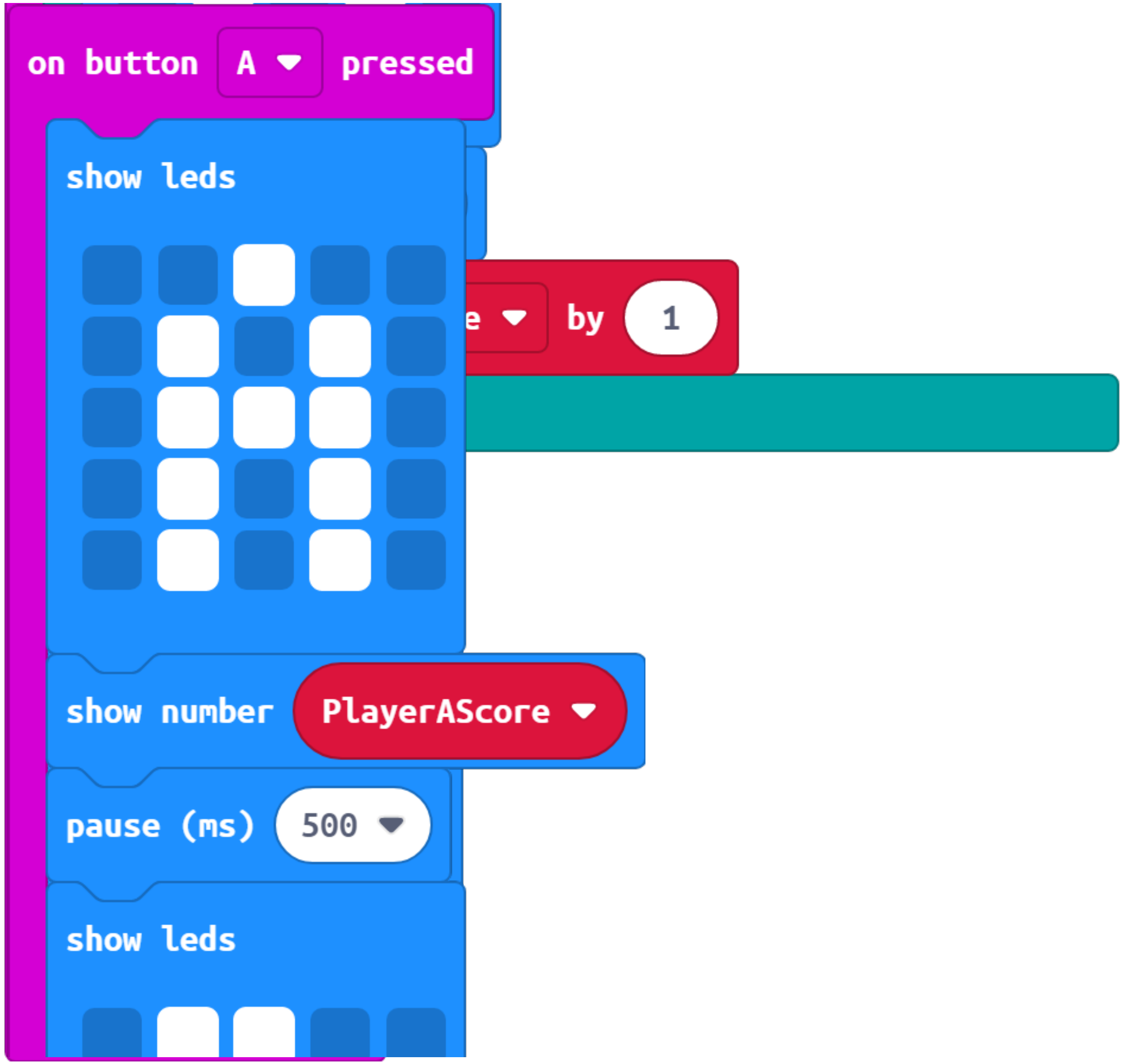
Your final program might look like:



Solution link
makecode.microbit.org/_YHuAxKere6vM

Your final code

Your final program might look like:



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Your final code

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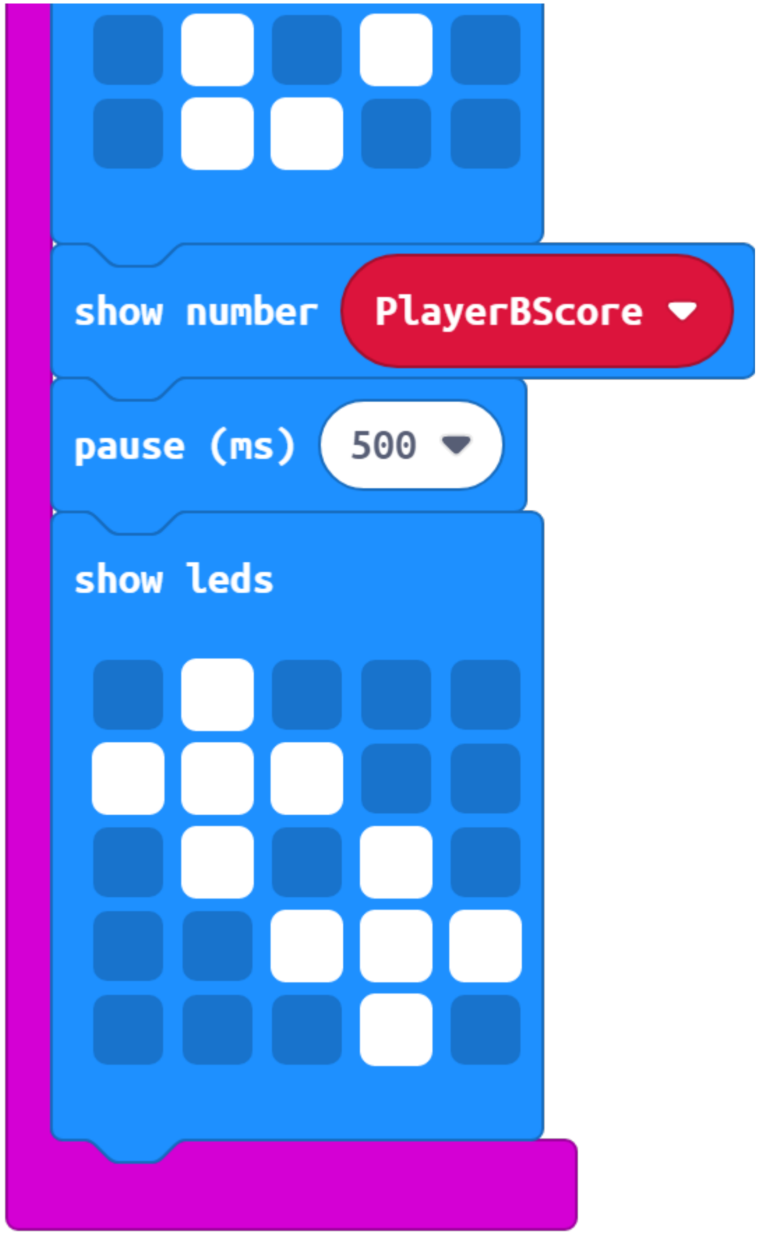

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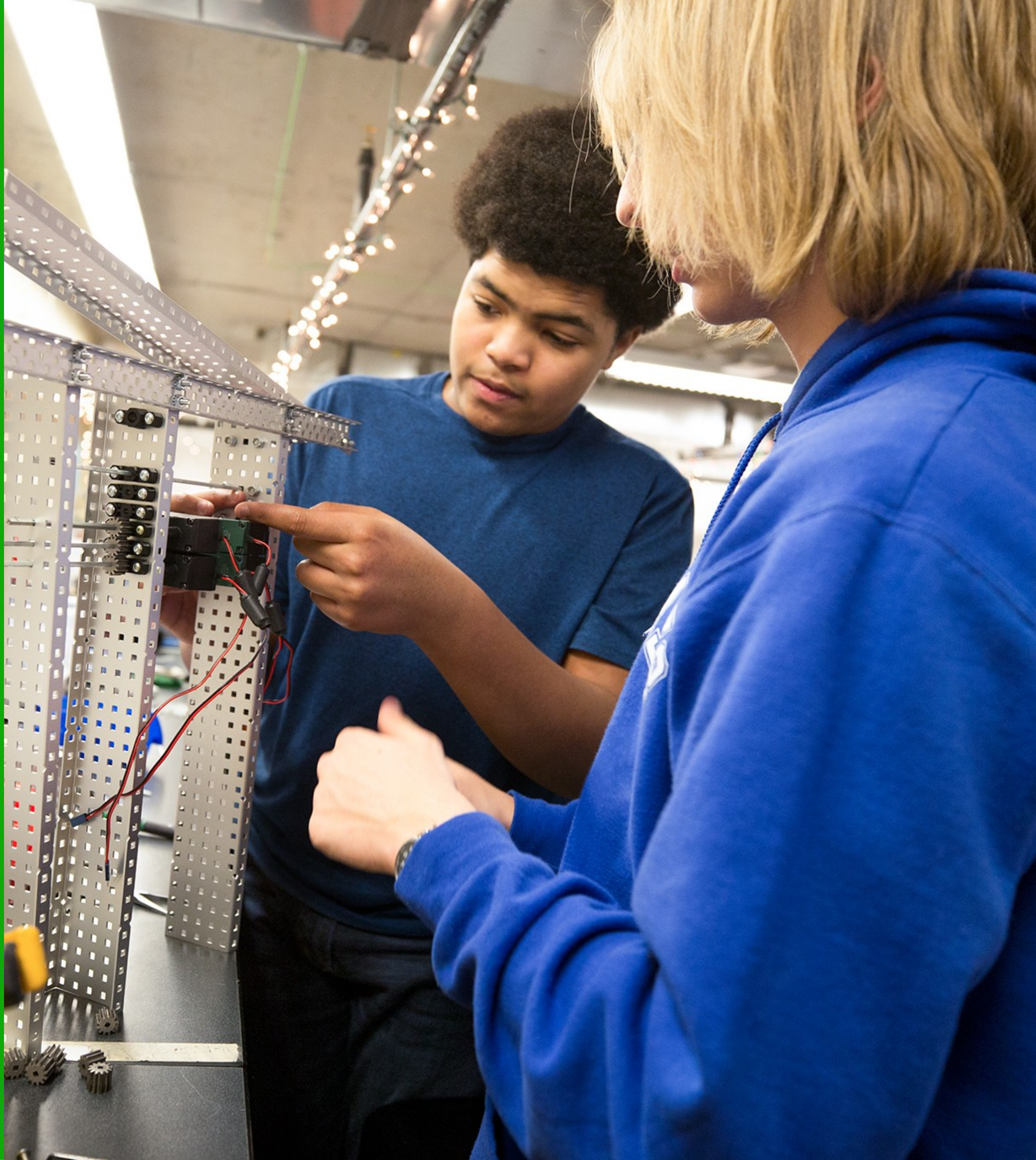
Time for a quiz!

What we learned in Lesson B

- Use Booleans to code a random double coin flipper
- How much we’ve already learned!

Next time: Lesson C

- Project Boolean



Lesson C
Project Boolean

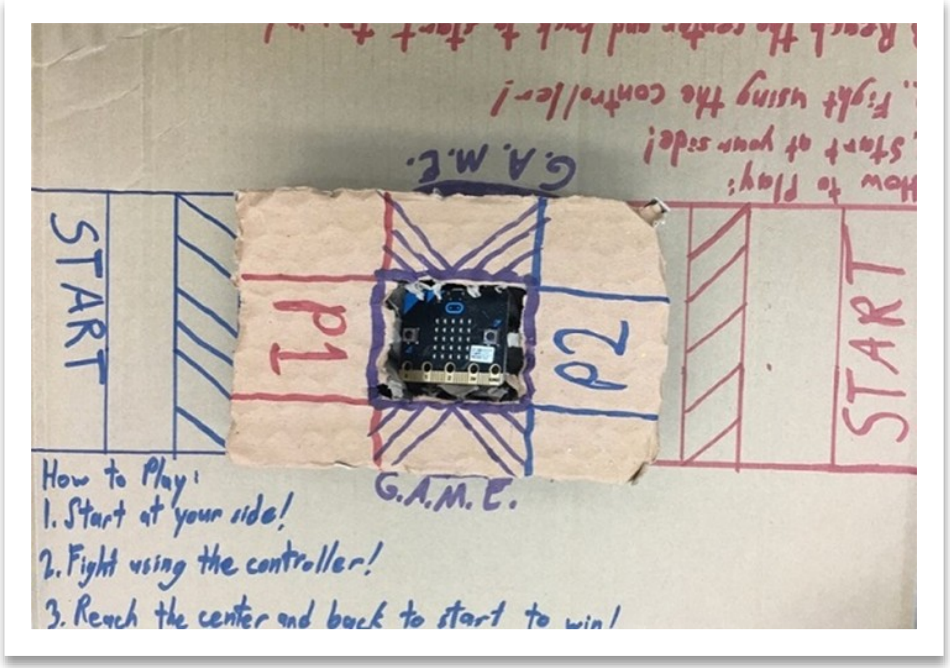
What we'll learn in Lesson C

- Code a micro:bit program that uses Boolean variables and Boolean operators

Activity: Project Boolean

Create a micro:bit program that uses more than two Boolean values:

- Boolean variables
- Boolean operators
- Possibly the random function



Available inputs:

- Acceleration
- Light level
- Rotation
- Button is pressed
- Compass heading
- Temperature
- Running time
- On shake
- On button pressed
- On logo down
- On logo up
- On pin pressed
- On screen down
- On screen up
- Pin is pressed

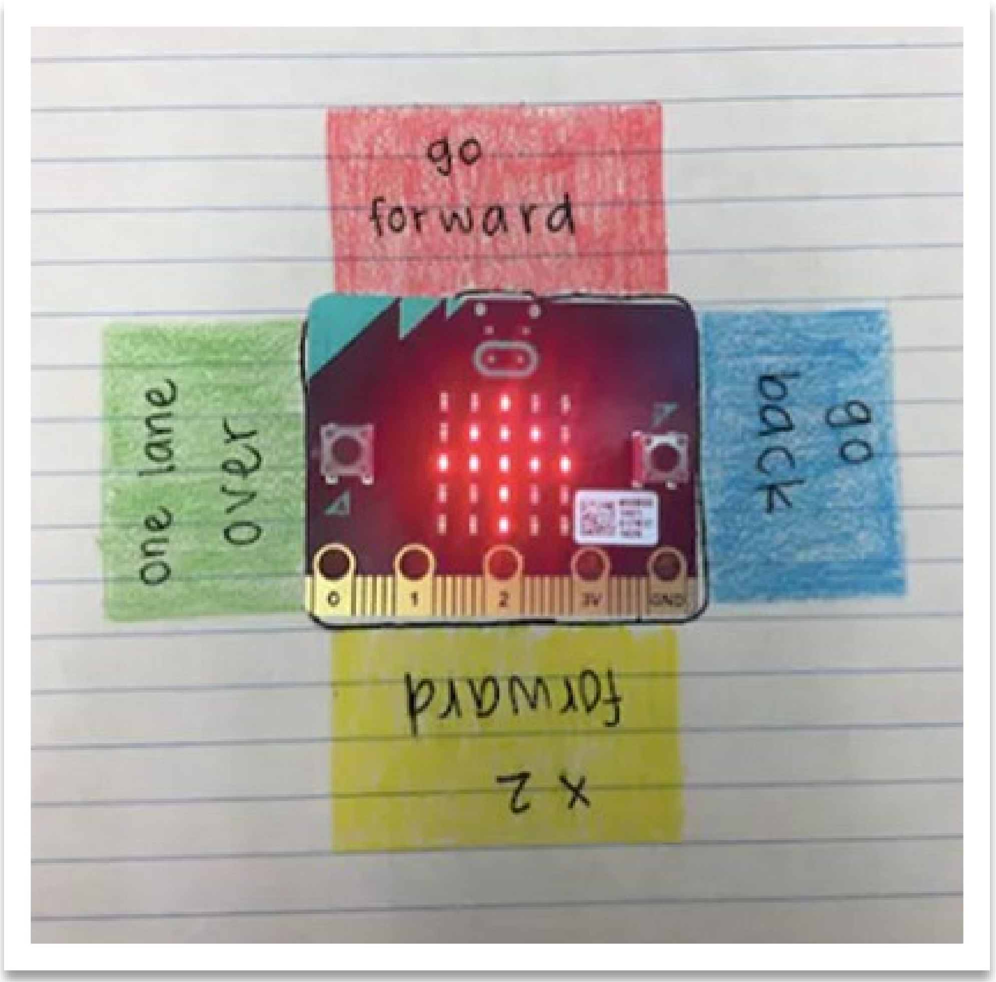
Project idea: Sunscreen monitor

- When you shake the micro:bit, it reports the current temperature in degrees Fahrenheit.
- **Button B** measures the light, and if it is above 70 degrees AND bright, it'll show a sun icon.
- If it is above 70 degrees and less bright, it will display a cloudy symbol.
- If it is dark, it will display a nighttime icon.

NEW
Neutrogena

Project idea: Two-player game

- Create a game in which two players take turns on the same micro:bit.
- Use a Boolean variable called **PlayerATurn** to keep track of whose turn it is.
- Use Boolean variables and random values as part of a board game.
- Make a board, pieces, and a holder for the micro:bit.
- Try modding a current board game.



Project expectations

Use a design thinking approach and ensure the project meets the following specifications:

- **More than two Boolean variables** are implemented in a meaningful way
- The micro:bit program uses **Booleans** in a way that is **integral** to the program
- The program **compiles and runs as intended** and includes **meaningful comments** in code
- Provide the written **Reflection Diary** entry

Reflection Diary

- How did you incorporate Boolean variables into your micro:bit program?
- How did you incorporate Boolean operators into your micro:bit program?
- Describe something in your project that you are proud of.
- If you had more time to work on this project,
  describe what you might add or change.
- Publish your MakeCode program and include the link.

What we learned in Lesson C

- Coded a micro:bit program using Boolean variables and Boolean operators